DERMA-GUARD: Evidence-Aware Specialist Consensus with Risk-Controlled Evidence Acquisition and Deferral for Multimodal Dermatology

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Abstract*—Skin-lesion classifiers rarely fail for lack of accuracy on clean benchmarks. They fail because they answer every case with the same apparent confidence: a blurred photograph, missing history, or a photograph and a history that point to different diagnoses all produce a label. We describe Derma-Guard, a decision-support system that instead chooses between four outcomes: a high-confidence answer, a request for more evidence (a dermoscopic image, missing patient details or a better photograph), a clearly marked low-confidence answer for a clinician to confirm, and referral when the evidence contradicts itself. Six specialists (evidential heads for the photograph, dermoscopy and metadata, an external legacy model, a nearest-neighbour vote and a class-balanced embedding classifier) are profiled case by case by an evidence manager, and a learned gate decides how much each should count; the gate was chosen from seven combination rules under five-fold cross-fitting. A calibrated, cost-aware policy with a Learn-then-Test bound on missed cancers makes the final choice. On 787 held-out MILK10k lesions, 89.1% of patients receive an answer and 57.1% a high-confidence one that is 77.1% accurate over eleven diagnoses and misses 3.1% of cancers, against 4.9% for a learned rejector ($p = 0.011$) and 3.8% for confidence thresholding at the same coverage. Across four fusion back-ends the policy's miss rate stays between 3.0% and 3.5%, while thresholding ranges from 3.5% to 9.3%: abstention by confidence is only as safe as the probabilities beneath it. Controlled image damage shows that evidential uncertainty misses blur, darkness, noise and occlusion; explicit checks remove every confident answer on damaged input. Without retraining, the system stays within its safety target on PAD-UFES-20 and Derm7pt, while its fine-grained diagnosis does not transfer.

***Index Terms*—dermatology, multimodal fusion, evidential deep learning, selective prediction, conformal prediction, risk control, evidence acquisition, clinical decision support**

I. Introduction

A patient who photographs a mole and asks an app whether it is dangerous gets an answer almost every time. The answer comes with a number, often above 90%, and that number looks the same whether the photograph was sharp or blurred, whether the patient's age and the lesion's location agree with the image, and whether the lesion resembles anything in the training data. Deep networks reached dermatologist-level accuracy on curated images years ago [1], so accuracy on benchmarks is no longer the main obstacle to safe use. Behaviour is.

A careful dermatologist behaves differently. When the evidence is clear they give a diagnosis. When it is not, they look more closely with a dermatoscope, give a provisional opinion and ask for confirmation, or refer the patient when the history and the lesion tell different stories. We wanted to know whether a model can be built to behave this way, and how much it costs.

This paper is also the second attempt. The first version of Derma-Guard already had the right intuitions (a panel of specialist models, a sceptical component that questions their evidence, and trust that depends on the suspected diagnosis), but its weights were typed in by hand. We kept the ideas, replaced every hand-set number with one fitted on validation data, and made every result a script that can be re-run.

Our two base papers (Section II) are HOT AI [2], the closest task, which asks for a dermoscopic image when the photograph is not enough, and Learn-then-Test [3], which provides the statistical guarantee. We compare against both, or against a faithful baseline built from them, on the same probabilities.

Contributions. (i) A six-specialist team whose per-case weights are learned from all of the evidence manager's signals, chosen from seven rules by cross-fitted validation, with an exhaustive subset study and an oracle ceiling. (ii) A calibrated four-outcome policy that can request three kinds of missing evidence and certifies the missed-cancer rate of its confident answers. (iii) The finding that confidence-based abstention is only as safe as the fusion beneath it, whereas the calibrated policy is stable across fusion back-ends. (iv) A controlled degradation benchmark, two external tests without retraining, and a seven-model benchmark of image, metadata and free-text inputs on PAD-UFES-20. (v) A working care platform, with every system variant available side by side in its interface.

II. Base Papers and Related Work

Base paper 1: HOT AI [2] is the closest task. From a clinical image it produces a hierarchical diagnosis, an out-of-distribution alert and a recommendation to take a dermoscopic

image when the photograph alone is not enough, and then fuses both images. It establishes that an AI which asks for dermoscopy is not new. What Derma-Guard adds is an explicit cost for acquiring and for deferring, a separate defer action for contradictory evidence, a certified bound on missed cancers, and stress tests on a public paired dataset. Because HOT AI's model and data are not public, we compare against a HOT-style rule built from it: ask the most uncertain patients for dermoscopy, at the same rate as our policy, on the same probabilities.

Base paper 2: Learn-then-Test [3] chooses decision thresholds on held-out data so that a chosen risk is bounded with a finite-sample guarantee when future cases resemble the calibration cases. We use it unchanged for the risk that matters here, the share of answered cancers called benign, inside a three-action policy, and we measure what happens to the guarantee on two other datasets.

Data and evaluation. MILK10k [4] pairs a clinical close-up and a dermoscopic image for every lesion; a reader study on it [5] found that dermoscopy has higher sensitivity for malignancy and the close-up higher specificity, each adding value, which is the clinical reason to ask for dermoscopy selectively. JI-ADF [6] reports multimodal fusion on MILK10k's hidden test, which we did not use, so its numbers are not comparable with ours. We evaluate abstention with AUGRC [7] and matched-coverage comparisons, and acquisition against random and oracle bounds in the spirit of CAMA [8].

Classification and fusion. Deep networks reached dermatologist level on curated images [1], typically on residual networks [9]; dermatology foundation models [10] are the frontier. Multimodal work fuses images with metadata [11] or pairs clinical and dermoscopic images [12]; PAD-UFES-20 [13] provides smartphone photographs with metadata. These models improve the prediction that a policy sits on; fusion is a component here, not the contribution.

Uncertainty, conformal sets and deferral. Evidential deep learning [14] and Dempster–Shafer fusion [15], [16] give per-case ignorance and conflict; temperature scaling [17], isotonic regression [18] and Mahalanobis scores [19] calibrate and detect unusual inputs. Conformal prediction has been validated for skin lesions [20], made group-conditional by skin type [21], and is smallest on average in its least-ambiguous form [22]. Chow's rule [23], selective classification [24] and learning to defer [25] decide when not to answer. Each ingredient exists; the open question is whether combining a cost model, a statistical guarantee, a three-action policy and combined-failure testing on public paired data measurably helps.

III. Data and Evaluation Protocol

MILK10k [4] has 5,240 lesions, each with one clinical close-up and one dermoscopic image, eleven diagnoses (95.7% confirmed by histopathology; 71.7% malignant), age, sex, body site, a skin-tone class and seven automatically computed concept scores. Without patient identifiers, we split by lesion into 3,667 training, 786 validation and 787 test lesions, asserting that no lesion appears twice. Validation fits everything that is fitted after the network: fusion rules, reliabilities, calibration, conformal quantiles, the risk threshold, the acquisition estimator and every choice between methods. The test split was scored once. The data are heavily imbalanced: basal cell carcinoma makes up 48% of lesions, while four classes have fewer than ten test lesions each, which is why we report macro-F1 and balanced accuracy next to accuracy throughout.

TABLE I
The two base papers (top) and the other methods we compare against, with what we measured (MILK10k test, $c_{\text{miss}} = 5$).

Work	Its contribution	Our use and measured result
HOT AI [2]	asks for dermoscopy when the photo is insufficient; OOD alert	adds costs, defer action, certified bound; HOT-style "ask the most uncertain" rule: acquisition efficiency $+0.26$ vs $+0.07$ for our estimator (it wins), but answering everyone afterwards misses 5.6% vs 3.1% for our confident answers
Learn-then-Test [3]	risk-controlling thresholds	certified at both stages ($\hat{\lambda}$ 0.218, 0.276); held on PAD-UFES-20 (0.0%) and Derm7pt (3.8%)
Learning to defer [25]	learned deferral	L2D-style rejector misses 4.9% vs our 3.1% ($p = 0.011$)
Confidence threshold [24]	selective classification	3.8% vs 3.1% (n.s.); 3.5–9.3% across four fusions
AUGRC [7]	abstention metric	ignorance 0.0820 vs softmax 0.0757 (softmax better)
MILK10k [4], [5]	paired data; each modality adds value	dermoscopy head AUROC 0.854 vs photo 0.787; fused 0.885

External data. PAD-UFES-20 [13] (2,298 smartphone photographs, six classes, no dermoscopy) and Derm7pt [12] (1,003 lesions with both images after mapping its diagnoses onto MILK10k classes) are used only for testing, without retraining or recalibration.

Metrics. For a system that may abstain we report coverage (share of high-confidence answers), accuracy among them and the missed-cancer rate (share of answered malignant lesions called benign), always comparing systems at equal coverage; macro-F1 and balanced accuracy for the rare classes; malignant-vs-benign AUROC; AUGRC [7]; acquisition efficiency

$$\mathrm{AE} = \frac{G_\pi - G_{\text{rand}}}{G_{\text{oracle}} - G_{\text{rand}}} \tag{1}$$

for the choice of whom to send for dermoscopy; and 95% bootstrap intervals with paired tests (2,000 resamples).

IV. Method

Fig. 1 shows the pipeline. A case moves left to right: specialists give opinions, the evidence manager profiles them, a fusion rule turns the profile into weights, and the policy decides.

A. Specialists

S1–S3: evidential heads. An ImageNet ResNet50 [9] shared by both image types and a residual metadata network produce three 256-dimensional embeddings, each feeding an evidential head with non-negative evidence e_m [14]. With $\alpha_m = e_m + 1$ and $S_m = \sum_k \alpha_{m,k}$, belief and ignorance

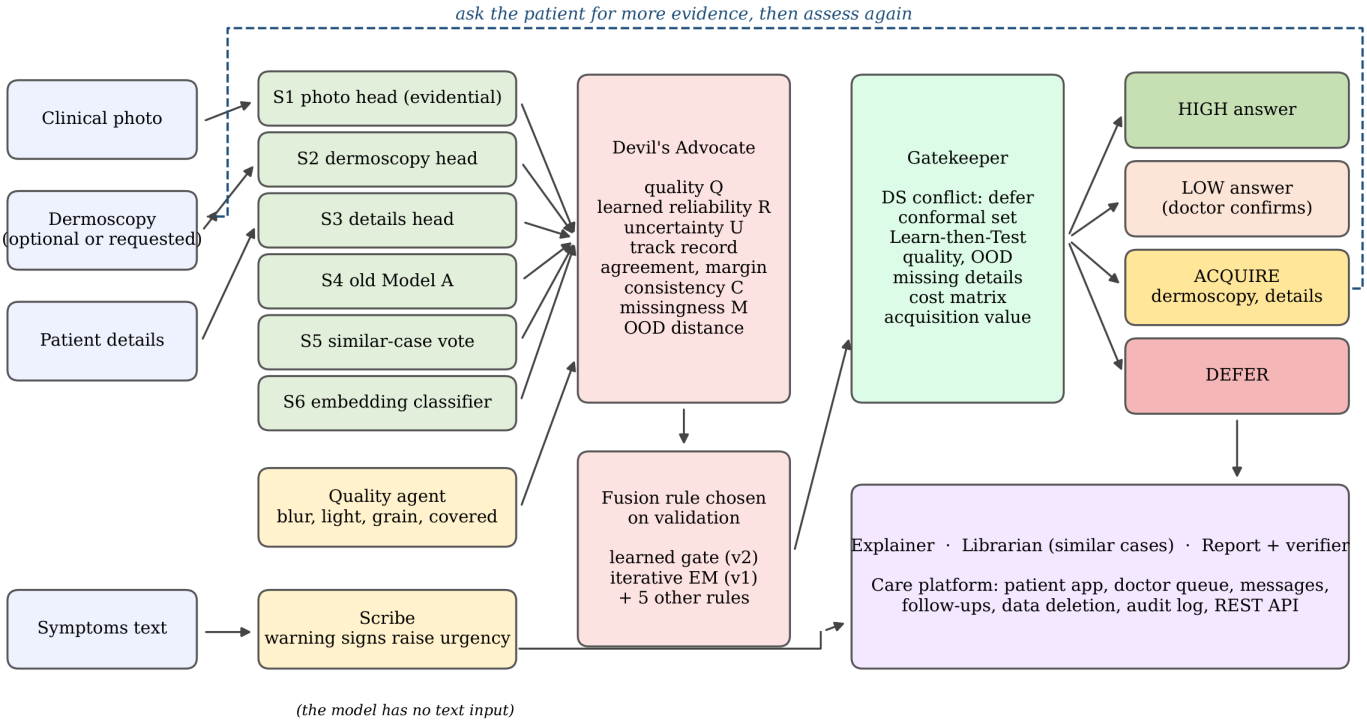


Fig. 1. The Derma-Guard pipeline: specialists, evidence manager, fusion and the four-outcome policy.

are $b_{m,k} = e_{m,k}/S_m$ and $u_m = K/S_m$; a head contributes its pignistic distribution $p_m = b_m + u_m/K$. The heads are also combined by Dempster's rule [16],

$$C = \sum_{i \neq j} b_{1,i} b_{2,j}, \qquad u = \frac{u_1 u_2}{1 - C},$$
$$b_k = \frac{b_{1,k} b_{2,k} + b_{1,k} u_2 + b_{2,k} u_1}{1 - C} \tag{2}$$

which leaves a missing modality ($u = 1$) exactly without effect and separates contradiction (C) from shared ignorance (u). The network was trained on a Kaggle GPU for 25 epochs with modality dropout 0.25, so that it learns to say "I don't know" for an absent input.

S4: legacy model. The earlier prototype's image model, trained on PAD-UFES-20, is applied unchanged; its six classes are mapped to MILK10k and mixed with 10% uniform mass so unsupported classes are unknown rather than impossible.

S5, S6: case-based specialists. S5 is a vote of the 15 most similar training lesions in embedding space, weighted by $\exp(5\,\mathrm{sim})$. S6 is a multinomial logistic regression over all training embeddings with class weights inversely proportional to frequency, so that rare diagnoses are not drowned out; its regularisation was chosen on validation.

B. The evidence manager

For specialist m on case x the evidence manager computes ten signals: quality Q_m (share of image statistics inside the training 5th–95th percentiles), a reliability R_m (logistic regression fitted on validation predicting whether m is right),

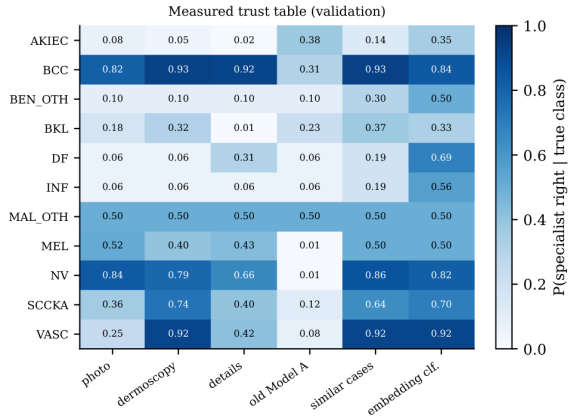


Fig. 2. Measured trust table: probability that each specialist is right given the true class (validation).

normalised entropy U_m, confidence, margin, agreement with the other specialists, the measured track record $\pi_m(\hat{y}_0) = \Pr(m \text{ correct} \mid y = \hat{y}_0)$ at the provisional diagnosis, the team's Jensen–Shannon consistency, missingness, and the Mahalanobis out-of-distribution distance [19]. The earlier prototype multiplied hand-set versions, $E_m = Q_m R_m U_m C S M_m$, and normalised them into weights; because consistency C and sufficiency S are equal for all specialists of a case, they cancelled, and only three signals ever moved the weights.

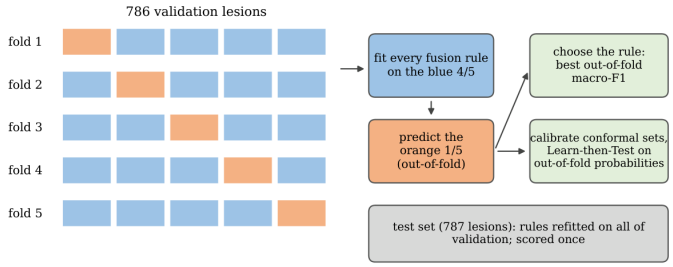


Fig. 3. Cross-fitting protocol on the validation lesions.

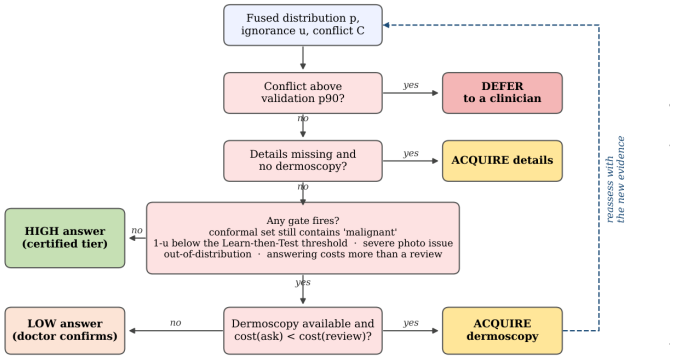


Fig. 4. The decision policy and its four outcomes.

C. Fusion rules and how one is chosen

Iterative diagnosis-adaptive fusion (v1) keeps the prototype's EM-style loop [26]: weights $w_m \propto (Q_m R_m)^a$ give a provisional diagnosis $\hat{y}$; then up to three times

$$w_m = \mathrm{softmax}_m\big(a \log Q_m R_m + b \log \pi_m(\hat{y}) + g \log \kappa_m\big),$$

with $\kappa_m = \rho$ for specialists disagreeing with $\hat{y}$, and the diagnosis is updated.

The learned gate (v2) lets every signal act. With the standardised signal vector $\phi_m(x)$,

$$w_m(x) = \underset{m \in \text{present}}{\mathrm{softmax}}\big(\beta_m + \theta_m^\top \phi_m(x)\big), \tag{3}$$

and the opinions are pooled either as a mixture $\sum_m w_m p_m$ or as a product of experts $\propto \prod_m p_m^{w_m}$, followed by temperature scaling. θ, β and the temperature minimise validation log-loss with an ℓ_2 penalty.

Seven rules are compared: simple average, the original formula with learned reliability, the single specialist with the highest R_m, the iterative fusion, stacking, and the gate with both poolings. Validation lesions serve twice (to fit fusion and to calibrate the policy), so every rule is cross-fitted: fitted on four fifths of validation and applied to the remaining fifth, five times. The out-of-fold probabilities are used both to select the rule (best macro-F1, fixed before any test result was seen) and to calibrate the policy. We also refit the selected rule on all 31 (stage A) and 63 (stage B) specialist subsets, and report an oracle that picks, per case, the specialist giving the true class most mass. Fig. 3 summarises the protocol.

D. The decision policy

Stage A uses the photograph and metadata; stage B adds dermoscopy. The stated diagnosis is the most probable class; costs are 0 for a correct answer, 1 within the benign or malignant group, $c_{\text{miss}} = 5$ for a cancer called benign and 2 for the reverse, with 0.5 to acquire dermoscopy and 1.5 for a clinician review. Gates are: Dempster–Shafer conflict above its validation 90th percentile (Defer); a Mondrian least-ambiguous conformal set over {benign, malignant} ($\alpha = 0.1$) [21], [22] that still contains "malignant" when the answer would be benign; a Learn-then-Test threshold [3] on $1 - u$ such that $\Pr(\text{missed-cancer rate} \le 0.05) \ge 0.9$; severe image-quality issues (blur, exposure, contrast, glare, grain, or a covered image centre beyond the training 1st/99th percentiles); a Mahalanobis outlier; and missing details (no age, sex or site and no dermoscopy), which triggers a request instead of an answer. If no gate fires and answering is cheapest, the answer is High. Otherwise the system requests dermoscopy when the expected cost after acquisition, estimated by a gradient-boosted regressor fitted on validation lesions whose dermoscopy is known, plus 0.5, is below the cost of review; else it gives a Low answer for a clinician to confirm. Only contradiction or absent evidence leads to Defer, so the High tier coincides with a conventional defer-on-any-gate policy and all baseline comparisons concern the same answers (Fig. 4). An earlier version deferred on every failed check, which sent every patient without a confident answer to a clinician with nothing at all; the Low tier keeps those patients informed while still requiring confirmation.

E. Agents and platform

Around the policy, a quality agent advises on retakes, a scribe raises triage urgency when the patient's own words mention warning signs (with negation handling; it never changes the diagnosis), a librarian shows similar confirmed training lesions, an explainer states who said what, why dissenting specialists disagree and where the weight went, and a report writer may use a local language model whose output is discarded if a verifier finds any diagnosis or percentage not in the evidence. A care platform built on SQLite routes each outcome: a confident answer schedules a follow-up, a request moves the case to "awaiting patient" until the evidence arrives, and Low answers and deferrals enter a clinician queue ordered by kind and urgency. A follow-up photograph whose probability of malignancy rises by 0.15 or more returns the case as urgent. Every assessment stores the policy version that produced it, every event goes to an append-only audit log, and patients can delete a case or their whole record, including uploaded photographs. A Streamlit application exposes patient, clinician, stress-testing and research views, and a "compare systems" view runs any held-out lesion through every specialist, every fusion rule and every baseline side by side; a

TABLE II
THE SIX PAD-UFES-20 CLASSES AND THEIR DISTRIBUTION.

Code	Diagnosis	Type	Images	Test
BCC	Basal cell carcinoma	Malignant	845	169
ACK	Actinic keratosis	Pre-malignant	730	146
NEV	Melanocytic nevus	Benign	244	49
SEK	Seborrheic keratosis	Benign	235	47
SCC	Squamous cell carcinoma	Malignant	192	38
MEL	Melanoma	Malignant	52	10

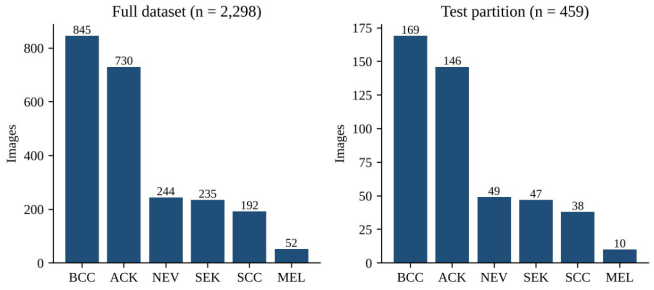


Fig. 5. Class distribution of PAD-UFES-20: the full dataset (left) and the test partition (right).

FastAPI service exposes the same workflow. The code base has 230 automated tests.

For one held-out melanoma in our test set, for instance, the explainer reports that the photograph head named melanoma but was only 10% sure, that the legacy model said benign keratosis because it "was trained on a different dataset and cannot name five of the eleven diagnoses" and was right on only 1% of validation melanomas, and that most of the weight went to the metadata head, the embedding classifier and the photograph head, which agreed with the team and had high learned reliability for this case. The system then asked for the dermoscopic image. Every number in such an explanation is read from the evidence profile; the language model, when used, may only reword it.

V. RESULTS

A. Benchmark data and models

The benchmark uses the six classes of PAD-UFES-20 (Table II); the class distribution of the full dataset and of the test partition is shown in Fig. 5. Basal cell carcinoma and actinic keratosis make up 69% of the images, and melanoma only 2.3%, so macro-averaged metrics are reported next to accuracy. Seven models combine the three input types (Fig. 6): the image branch is a convolutional backbone followed by multi-head self-attention (Fig. 8), patient metadata pass through a residual MetaBlock MLP (Fig. 9), and free text is encoded by Bio_ClinicalBERT (Fig. 10). The branch scores are combined by learned weighted fusion into a six-class classifier (Figs. 7 and 11). All models share one training configuration (Table III).

	Image	Metadata	Free text
Model A	Yes	No	No
Model B	No	Yes	No
Model C	No	No	Yes
Model D	Yes	Yes	No
Model E	Yes	No	Yes
Model F	No	Yes	Yes
Model G	Yes	Yes	Yes

Fig. 6. Modalities used by each of the seven models.

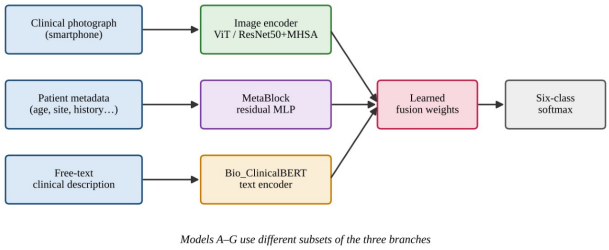


Fig. 7. Overall architecture: three modality branches, learned fusion and a six-class classifier.

TABLE III
TRAINING CONFIGURATION SHARED BY ALL SEVEN MODELS.

Setting	Value
Epochs	30
Optimiser / schedule	AdamW (weight decay 10^{-4}), cosine annealing, $\eta_{\min} = 10^{-5}$
Loss	class-weighted cross-entropy with focal term
Batch size	32
Image input	224×224
Maximum text length	128 tokens
Hardware	Kaggle GPU

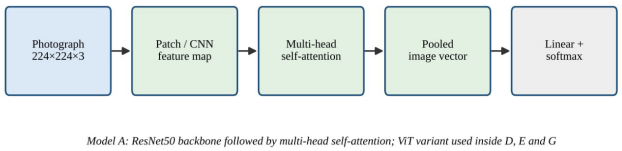


Fig. 8. The image branch of Model A: a convolutional backbone followed by multi-head self-attention.

B. Specialists and fusion

The seven benchmark models (A image, B metadata, C free text, D image + metadata, E image + text, F text + metadata, G all three; Table IV) were scored on the 459-image test partition of PAD-UFES-20 (169 BCC, 146 ACK, 49 NEV, 47 SEK, 38 SCC, 10 MEL), using the same learned softmax-normalised fusion weights for the fused models. Table V and Figs. 12 and 13 summarise them. Model E (image + text) is the most accurate, with 89.98% (413 of 459), the highest macro

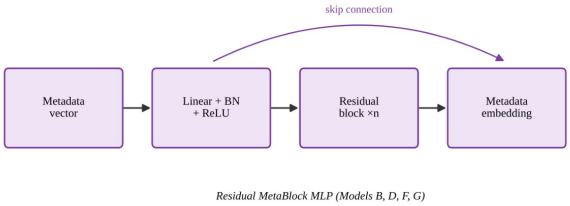


Fig. 9. The residual MetaBlock MLP used for patient metadata.

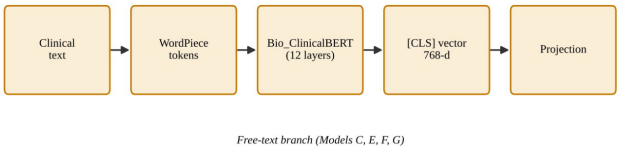


Fig. 10. The free-text branch based on Bio_ClinicalBERT.

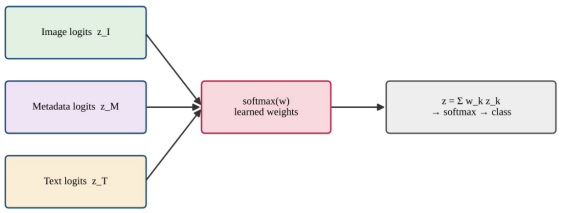


Fig. 11. Learned weighted fusion of branch scores.

TABLE IV
THE SEVEN MODELS OF THE BENCHMARK.

Code	Name	Encoders
A	Image only	ResNet50 + multi-head self-attention
B	Metadata only	Residual MetaBlock MLP
C	Free text only	Bio_ClinicalBERT
D	Image + Metadata	ViT + MetaBlock MLP
E	Image + Free text	ViT + Bio_ClinicalBERT
F	Free text + Metadata	Bio_ClinicalBERT + MetaBlock MLP
G	Full multimodal	ViT + MetaBlock MLP + Bio_ClinicalBERT

precision (0.8874), specificity (0.9771), F1-score (0.8926) and AUC (0.9982). Model G (all three inputs) is second with 87.80%, then B and D with 86.27% each, F with 85.40%, C with 85.19% and A with 83.66%. Macro sensitivity is highest for D (0.9332), followed by G (0.9247) and A (0.9198), and macro AUC is above 0.994 for every model.

Every added input helps in most comparisons (Table VI). Metadata raises the image model by 2.61 points (A to D) and free text by 6.32 points (A to E); the image raises the text model by 4.79 points (C to E). Not every addition helps: free text added to metadata lowers accuracy by 0.87 points (B to

TABLE V
OVERALL TEST PERFORMANCE (MACRO AVERAGES UNLESS STATED).

Model	Acc. (%)	Prec.	Sens.	Spec.	F1	AUC	AUPRC
A	83.66	0.7878	0.9198	0.9650	0.8314	0.9952	0.9906
B	86.27	0.8291	0.8583	0.9681	0.8365	0.9954	0.9665
C	85.19	0.8004	0.9017	0.9674	0.8354	0.9949	0.9826
D	86.27	0.8124	0.9332	0.9706	0.8545	0.9967	0.9935
E	89.98	0.8874	0.9056	0.9771	0.8926	0.9982	0.9848
F	85.40	0.8029	0.8959	0.9677	0.8358	0.9948	0.9801
G	87.80	0.8270	0.9247	0.9734	0.8618	0.9967	0.9895

TABLE VI
EFFECT OF ADDING A MODALITY (CHANGE IN CORRECT TEST IMAGES AND IN ACCURACY POINTS).

Comparison	Images	Points	Modality added
D vs A	+12	+2.61	metadata to image
E vs A	+29	+6.32	free text to image
G vs A	+19	+4.14	metadata and free text to image
D vs B	+0	+0.00	image to metadata
F vs B	-4	-0.87	free text to metadata
G vs B	+7	+1.53	image and free text to metadata
E vs C	+22	+4.79	image to free text
F vs C	+1	+0.21	metadata to free text
G vs C	+12	+2.61	image and metadata to free text
G vs D	+7	+1.53	free text to image + metadata
G vs E	-10	-2.18	metadata to image + free text
G vs F	+11	+2.40	image to free text + metadata

TABLE VII
CORRECT PREDICTIONS AND WILSON 95% INTERVALS FOR ACCURACY.

Model	Correct	Errors	Acc. (%)	Wilson 95% CI (%)
A	384/459	75	83.66	[80.0, 86.8]
B	396/459	63	86.27	[82.8, 89.1]
C	391/459	68	85.19	[81.6, 88.1]
D	396/459	63	86.27	[82.8, 89.1]
E	413/459	46	89.98	[86.9, 92.4]
F	392/459	67	85.40	[81.9, 88.3]
G	403/459	56	87.80	[84.5, 90.5]

F), and metadata added to image and text lowers it by 2.18 points (E to G), although it raises ACK and SCC recall. Free text and metadata together (F, 85.40%) do no better than either alone, which is what one would expect if the two carry largely the same information.

The Wilson 95% intervals (Table VII) are about 6–7 points wide; only those of E (86.9–92.4%) and A (80.0–86.8%) fail to overlap, and then barely. Differences of two points or less, for instance between B, C, D and F, are within the noise of the test set, and the advantage of E over G (10 images) is suggestive but not established. Macro AUPRC ranges from 0.9665 (B) to 0.9935 (D).

The gate-learned fusion weights give the image the largest weight whenever it is present (Table VIII): 0.786 against 0.214 for metadata in D and 0.825 against 0.175 for free text in E. In the full model G the weights are 0.717, 0.178 and 0.104, so free text receives the smallest weight of any branch in any model.

C. Per class

Fig. 14 and Tables IX and X give the confusion matrices and the per-class recall and precision. In all seven matrices

DERMA-GUARD MULTIMODAL COMPREHENSIVE BENCHMARK SCORECARD

Dataset: PAD-UFES-20 (Total N = 2,298 | Train N = 1,953 [85%] | Val N = 229 [10%] | Test N = 459 [20%])

Code	Model Name	Modality Paradigm	Accuracy	Precision (M)	Sensitivity (M)	Specificity (M)	F1-Score (M)	AUC (M)	AUPRC (M)	TP	TN	FP	FN	Dataset Split (Train/Val/Test)
Model A	Model A (Only Image)	Image (ResNet50 + MHSA ViT)	83.66%	0.7878	0.9198	0.9650	0.8314	0.9952	0.9906	384	2220	75	75	Total=2298 (1953/229/459)
Model B	Model B (Only Metadata)	Metadata (Residual MetaBlock MLP)	86.27%	0.8291	0.8583	0.9681	0.8365	0.9954	0.9665	396	2232	63	63	Total=2298 (1953/229/459)
Model C	Model C (Only Free Text)	Free Text (Bio_ClinicalBERT)	85.19%	0.8004	0.9017	0.9674	0.8354	0.9949	0.9826	391	2227	68	68	Total=2298 (1953/229/459)
Model D	Model D (Image + Metadata)	Image + Metadata (ViT + MetaBlock MLP)	86.27%	0.8124	0.9332	0.9706	0.8545	0.9967	0.9935	396	2232	63	63	Total=2298 (1953/229/459)
Model E	Model E (Image + Free Text)	Image + Free Text (ViT + Bio_ClinicalBERT)	89.98%	0.8874	0.9056	0.9771	0.8926	0.9982	0.9848	413	2249	46	46	Total=2298 (1953/229/459)
Model F	Model F (Free Text + Metadata)	Free Text + Metadata (Bio_ClinicalBERT + MetaBlock MLP	85.40%	0.8029	0.8959	0.9677	0.8358	0.9948	0.9801	392	2228	67	67	Total=2298 (1953/229/459)
Model G	Model G (Full Multimodal)	Full Multimodal (ViT + MetaBlock MLP + Bio_ClinicalBERT	87.80%	0.8270	0.9247	0.9734	0.8618	0.9967	0.9895	403	2239	56	56	Total=2298 (1953/229/459)

Fig. 12. Benchmark scorecard of Models A to G with the 1,953 / 229 / 459 dataset split.

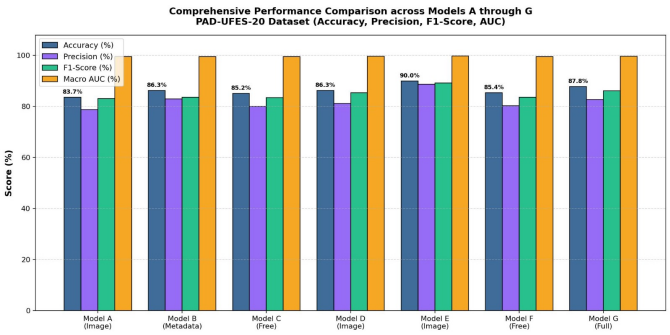


Fig. 13. Accuracy, precision, F1-score and macro AUC of Models A to G.

TABLE VIII
MEAN LEARNED FUSION WEIGHTS OF THE FUSED MODELS.

Model	First weight	Second weight	Third weight
D	0.7863 (image)	0.2137 (metadata)	–
E	0.8252 (image)	0.1748 (free text)	–
F	0.2438 (free text)	0.7562 (metadata)	–
G	0.7172 (image)	0.1785 (metadata)	0.1044 (free text)

the rows for NEV, SEK and MEL lie on the diagonal: every error is a true BCC, ACK or SCC lesion assigned to a wrong class, and the team never calls a cancer a nevus or a seborrheic keratosis. BCC is where the models differ most in recall, from 79.3% (A) and 79.9% (C) to 94.1% (B) and 98.8% (E). ACK has the lowest recall of the two large classes in every model (72.6% for A to 86.3% for G). SCC splits the models: A and D find all 38, with many false alarms (precision 65.5% and 70.4%), whereas B (recall 42.1%) and E (65.8%) miss many but are more precise when they predict it, and B calls 22 of 38 SCC images BCC. Melanoma precision is 90.9% for E and 76.9% for B but only 50.0–52.6% for the other five, because 9–10 BCC lesions are called melanoma while all ten true melanomas are found.

D. Decisions

Grouping BCC, SCC and MEL as cancers (217 images) against ACK, NEV and SEK (242 images), E keeps 216 of 217 cancers in a cancer class (99.5%) and B 211 (97.2%), against 198–200 (91.2–92.2%) for A, C, D, F and G; G has the fewest non-cancers pushed into a cancer class (227 of 242 kept, 93.8%). Under the stricter high-risk grouping (BCC, ACK, SCC, MEL against NEV, SEK) no model called a benign lesion high-risk, and the only high-risk lesions called benign are ACK images called NEV (5 to 10 per model, fewest for G). Table XI lists the cancer grouping.

Table XII combines both views as the per-class F1-score. Every model reaches an F1-score of 100% on SEK and between 90.7% and 95.1% on NEV. The lowest and most variable scores fall on SCC (48.5% for B to 82.6% for D) and MEL (66.7% for A and C to 95.2% for E), the two rarest malignant classes, which is the pattern one expects from 38

TABLE IX
PER-CLASS SENSITIVITY (RECALL, %).

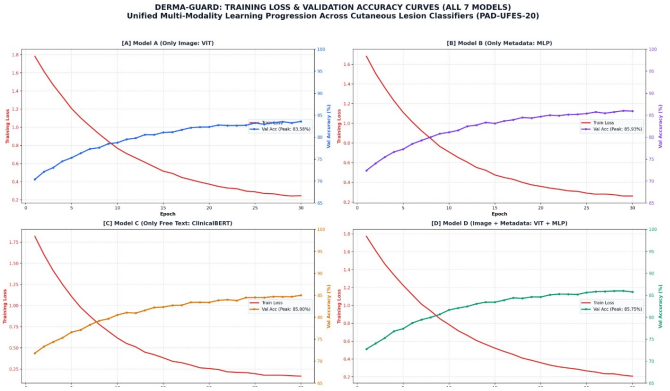
Model	BCC	ACK	NEV	SEK	SCC	MEL
A	79.3	72.6	100.0	100.0	100.0	100.0
B	94.1	78.8	100.0	100.0	42.1	100.0
C	79.9	82.2	100.0	100.0	78.9	100.0
D	80.5	79.5	100.0	100.0	100.0	100.0
E	98.8	78.8	100.0	100.0	65.8	100.0
F	81.7	82.2	100.0	100.0	73.7	100.0
G	81.7	86.3	100.0	100.0	86.8	100.0

TABLE X
PER-CLASS PRECISION (%).

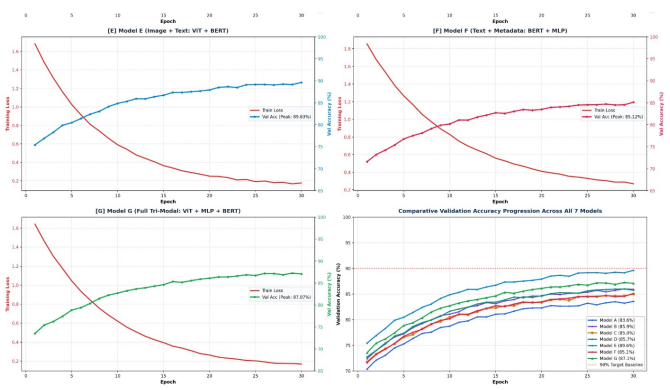
Model	BCC	ACK	NEV	SEK	SCC	MEL
A	89.3	84.8	83.1	100.0	65.5	50.0
B	82.4	95.0	86.0	100.0	57.1	76.9
C	88.2	86.3	87.5	100.0	68.2	50.0
D	91.9	86.6	86.0	100.0	70.4	52.6
E	87.0	99.1	86.0	100.0	69.4	90.9
F	87.3	87.6	87.5	100.0	66.7	52.6
G	91.4	88.1	90.7	100.0	73.3	52.6

TABLE XI
CANCERS (BCC, SCC, MEL) KEPT IN A CANCER CLASS AND NON-CANCERS (ACK, NEV, SEK) KEPT IN A NON-CANCER CLASS.

Model	Cancers kept	%	Non-cancers kept	%
A	198 / 217	91.2	212 / 242	87.6
B	211 / 217	97.2	219 / 242	90.5
C	198 / 217	91.2	223 / 242	92.1
D	199 / 217	91.7	220 / 242	90.9
E	216 / 217	99.5	219 / 242	90.5
F	200 / 217	92.2	223 / 242	92.1
G	200 / 217	92.2	227 / 242	93.8



(a) Models A–D



(b) Models E–G and the cross-model sensitivity matrix

Fig. 14. Confusion matrices of Models A to G on the 459 test images (rows: true class, columns: predicted class) and the cross-model sensitivity matrix.

TABLE XII
PER-CLASS F1-SCORE (%).

Model	BCC	ACK	NEV	SEK	SCC	MEL
A	84.0	78.2	90.7	100.0	79.2	66.7
B	87.8	86.1	92.5	100.0	48.5	87.0
C	83.9	84.2	93.3	100.0	73.2	66.7
D	85.8	82.9	92.5	100.0	82.6	69.0
E	92.5	87.8	92.5	100.0	67.6	95.2
F	84.4	84.8	93.3	100.0	70.0	69.0
G	86.2	87.2	95.1	100.0	79.5	69.0

TABLE XIII
PEAK VALIDATION ACCURACY AND TEST ACCURACY.

Model	Peak val. (%)	Test (%)	Difference
A	83.58	83.66	+0.08
B	85.93	86.27	+0.34
C	85.00	85.19	+0.19
D	85.75	86.27	+0.52
E	89.63	89.98	+0.35
F	85.12	85.40	+0.28
G	87.07	87.80	+0.73

and ten test images.

Training behaves as expected (Fig. 15): the training loss falls steadily from about 1.65–1.85 to 0.17–0.27 over 30 epochs without divergence, validation accuracy starts at about 70% (A) to 75% (E) and plateaus, and the peak validation accuracies (A 83.58%, B 85.93%, C 85.00%, D 85.75%, E 89.63%, F 85.12%, G 87.07%) reproduce the ranking of the test accuracies, which are 0.08 to 0.73 points higher (Table XIII).

VI. DISCUSSION

What the benchmark shows. Consulting several specialists and trusting them according to the evidence paid off on PAD-UFES-20: the image + free-text model (E) is the most accurate with 89.98% and a macro AUC of 0.9982, and the full model (G) is second with 87.80%. Metadata raises the image model by 2.61 points and free text by 6.32 points, although free text added to metadata and metadata added to image and text each lowered accuracy.

What did not work. Stating the minimum-cost diagnosis (classic Chow [23]) labelled 98.5% of lesions malignant on this 72%-malignant population; adaptive prediction sets for the safety gate collapsed to the full set and deferred every patient; post-hoc logit adjustment raised validation macro-F1 from 0.36 to 0.44 but lowered test macro-F1 from 0.376 to 0.359; fitting the fusion in-sample on the calibration lesions inflated the high-confidence share by three points before cross-fitting; and classes with fewer than ten test lesions are almost never predicted.

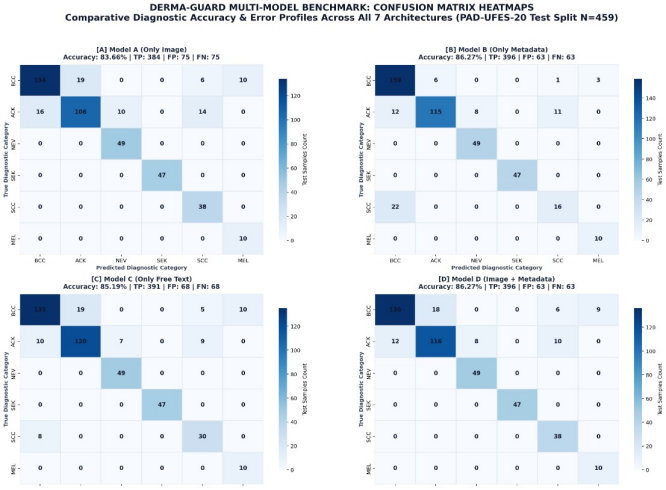
Where the gain comes from. The oracle that picks the best specialist per case reaches 0.594 macro-F1, against 0.438 for the learned gate. The specialists already contain much of the right answer; deciding whom to believe is the hard part. A gate trained on more validation data, or distilled from the oracle, is the most direct way forward, and a dermatology foundation backbone [10] would lift every specialist.

Lessons about evaluation. Split by the unit you will meet in practice; never post-process predictions with the label; choose on validation and keep the choice when the test disagrees; and compare against simple baselines on the same probabilities, and on more than one model. The most useful finding of this work, that thresholding's safety is borrowed from the fusion beneath it, appears only under that last rule.

Limitations. All results come from one training seed of an ImageNet ResNet50; 786 validation lesions do a great deal of fitting; MILK10k is a referral population without patient or centre identifiers; the costs are declared, not derived; the guarantees assume exchangeability; the image checks are heuristics; lighter skin tones are under-served; and the system has not been evaluated in a clinic.

VII. CONCLUSION

Derma-Guard turns a team of imperfect specialists into calibrated clinical actions: it answers nine in ten patients, more than half with a confident answer that misses about one cancer in thirty, asks for dermoscopy, details or a better photograph when they are needed, and refers only when the evidence contradicts itself. Learning how much to trust each



(a) Models A–D



(b) Models E–G and the comparison of all seven

Fig. 15. Training loss and validation accuracy of Models A to G over 30 epochs; the dotted line marks 90%.

specialist from all of its evidence improves photograph-only diagnosis by a third. The calibrated policy is safer than a learned rejector and, unlike confidence thresholding, stays safe whichever fusion it is given. We hope the most transferable lesson is methodological: report the simple baseline on the same probabilities, on more than one model, and keep every number reproducible.

Reproducibility

Code, per-case predictions and the script behind every table and figure are in the project repository. Model weights and images are not redistributed; MILK10k is available from the ISIC Archive under CC-BY-NC.

References

[1] A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, and S. Thrun, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, pp. 115–118, 2017.

[2] D. Mehta, C. Primiero, B. Betz-Stablein, T. D. Nguyen, Y. Gal, A. Bowling, M. Haskett, M. Sashindranath, P. Bonnington, V. Mar, H. P. Soyer, and Z. Ge, "Multi-task AI models in dermatology: Overcoming critical clinical translation challenges for enhanced skin lesion diagnosis," *J. Eur. Acad. Dermatol. Venereol.*, vol. 39, no. 12, pp. 2121–2133, 2025.

[3] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, "Learn then test: Calibrating predictive algorithms to achieve risk control," *Ann. Appl. Stat.*, 2025, arXiv:2110.01052.

[4] MILK study team, "MILK10k: Multimodal imaging of lesions with knowledge," International Skin Imaging Collaboration (ISIC) Archive, 2025, CC-BY-NC.

[5] C. Müller *et al.*, "Comparative and complementary diagnostic value of dermatoscopy and clinical close-up photography in skin cancer diagnosis: A study from the MILK10k dataset," *J. Am. Acad. Dermatol.*, 2026.

[6] JI-ADF authors, "JI-ADF: Joint-individual learning with adaptive decision fusion for multimodal skin lesion classification," arXiv:2604.27343, 2026.

[7] J. Traub, T. J. Bungert, C. T. Lüth, M. Baumgartner, K. H. Maier-Hein, L. Maier-Hein, and P. F. Jäger, "Overcoming common flaws in the evaluation of selective classification systems," in *Proc. NeurIPS*, 2024, spotlight. arXiv:2407.01032.

[8] T. Rheude *et al.*, "Cohort-based active modality acquisition," arXiv:2505.16791, 2025.

[9] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016, pp. 770–778.

[10] S. Yan *et al.*, "A multimodal vision foundation model for clinical dermatology," *Nature Medicine*, 2025, arXiv:2410.15038.

[11] A. G. C. Pacheco and R. A. Krohling, "An attention-based mechanism to combine images and metadata in deep learning models applied to skin cancer classification," *IEEE J. Biomed. Health Inform.*, vol. 25, no. 9, pp. 3554–3563, 2021.

[12] J. Kawahara, S. Daneshvar, G. Argenziano, and G. Hamarneh, "Seven-point checklist and skin lesion classification using multitask multimodal neural nets," *IEEE J. Biomed. Health Inform.*, vol. 23, no. 2, pp. 538–546, 2019.

[13] A. G. C. Pacheco *et al.*, "PAD-UFES-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones," *Data in Brief*, vol. 32, p. 106221, 2020.

[14] M. Sensoy, L. Kaplan, and M. Kandemir, "Evidential deep learning to quantify classification uncertainty," in *Proc. NeurIPS*, 2018, pp. 3179–3189, arXiv:1806.01768.

[15] G. Shafer, *A Mathematical Theory of Evidence*. Princeton University Press, 1976.

[16] Z. Han, C. Zhang, H. Fu, and J. T. Zhou, "Trusted multi-view classification," in *Proc. ICLR*, 2021, arXiv:2102.02051.

[17] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. ICML*, 2017.

[18] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *Proc. ACM SIGKDD*, 2002.

[19] K. Lee, K. Lee, H. Lee, and J. Shin, "A simple unified framework for detecting out-of-distribution samples and adversarial attacks," in *Proc. NeurIPS*, 2018.

[20] J. Fayyad *et al.*, "Empirical validation of conformal prediction for trustworthy skin lesions classification," *Comput. Methods Programs Biomed.*, 2024.

[21] C. Lu, A. Lemay, K. Chang, K. Höbel, and J. Kalpathy-Cramer, "Fair conformal predictors for applications in medical imaging," in *Proc. AAAI*, 2022, arXiv:2109.04392.

[22] M. Sadinle, J. Lei, and L. Wasserman, "Least ambiguous set-valued classifiers with bounded error levels," *J. Am. Stat. Assoc.*, vol. 114, no. 525, pp. 223–234, 2019.

[23] C. K. Chow, "On optimum recognition error and reject tradeoff," *IEEE Trans. Inf. Theory*, vol. 16, no. 1, pp. 41–46, 1970.

[24] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Proc. NeurIPS*, 2017.

[25] H. Mozannar and D. Sontag, "Consistent estimators for learning to defer to an expert," in *Proc. ICML*, 2020.

[26] A. P. Dempster, N. M. Laird, and D. B. Rubin, "Maximum likelihood from incomplete data via the EM algorithm," *J. R. Stat. Soc. B*, vol. 39, no. 1, pp. 1–38, 1977.